

Assignment 1: Basic Authentication

- Create a Spring Boot application with a REST API (`/hello` endpoint).
 - Secure it with **in-memory user authentication** (`user` / `password`).
 - Return different responses based on whether the user is authenticated or not.
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Assignment 2: Role-Based Access

- Add two roles: `ADMIN` and `USER`.
 - Create two APIs:
 - `/admin/dashboard` → accessible only to `ADMIN`.
 - `/user/profile` → accessible only to `USER`.
 - Try hitting these endpoints with different users.
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Assignment 3: Custom UserDetailsService

- Store users in a **database (MySQL)** instead of in-memory.
 - Implement `UserDetailsService` to fetch user credentials from the DB.
 - Use **BCrypt password encoder**.
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Assignment 4: JWT Authentication

- Convert the app to **stateless authentication** using JWT.

- Steps:
 - Implement `/login` API → generates JWT on valid credentials.
 - Add a JWT filter to validate tokens for every request.
 - Secure APIs (`/admin`, `/user`) with JWT instead of sessions.
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Assignment 5: Method-Level Security

- Enable method-level security (`@EnableGlobalMethodSecurity(prePostEnabled = true)`).
- Add a service with methods like:
 - `@PreAuthorize("hasRole('ADMIN')")` → `deleteUser()`
 - `@PreAuthorize("hasRole('USER')")` → `viewProfile()`
- Call these methods from controllers and test with different roles.